

2019-22

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers in
their own words as far as practicable.*

The figures in the margin indicate full marks.

Answer from both the Groups as directed.

Group – A

Answer any four questions : $10 \times 4 = 40$

- ✓1. What do you mean by Internet ? What are the advantages of internet ?
- ✓2. Write about the TCP/IP and also write its services. Also write about web client and web server.

3. What is javascript ? And discuss about javascript function with a program which performs addition operation between two given values.
4. What is javascript operator ? Write down about all javascript operators.
5. Write about XML and what are the differences between XML and SGML ? Also write about XML structuring data with an example.
6. What is DTD ? Also write about internal DTD and external DTD with examples.
7. What is JSP ? What are its life cycle ? Also write about JSP tags with an example.
8. Discuss about PHP, features of PHP and uses of PHP. What is string and write about some predefined string handling functions of PHP ?

Group – B

Answer all questions :

3×10 = 30

9. Write about domain name.

- ✓ 10. Write some disadvantages of internet.
11. What is type casting ? How type casting operation is performed in javascript ?
- ✓ 12. What is CSS ?
- ✓ 13. Discuss about XHTML.
- ✓ 14. What is HTML ?
15. Write about JSP comment tag.
16. What is static and dynamic web pages ?
- ✓ 17. What is the difference between compiler and interpreter ?
- ✓ 18. Write about the Web Browser.



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Answer any **four** questions of the following :

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1. What is an algorithm ? Explain the different characteristics of algorithm. Obtain the time complexity of the code given below :

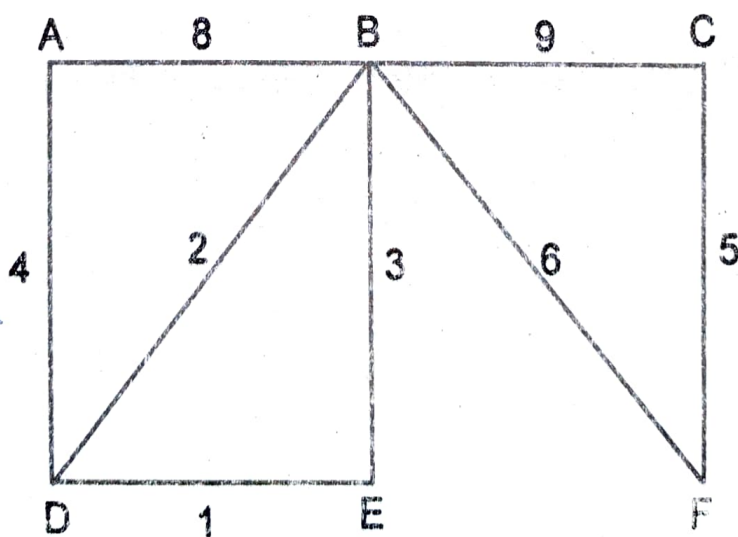
a = 0

for i = 1 to n do

for j = 1 to n – 1 do

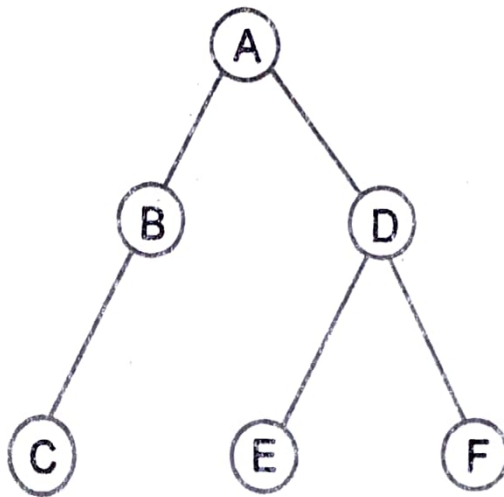
a = a + i * j;

2. What do you mean by Divide and Conquer Strategy? Sort the following list of elements using Quick sort. Also show intermediate steps of the operation. 29, 6, 27, 8, 6, 2, 45, 90.
3. Give the Binary search algorithm and analyze the efficiency.
4. Explain General Algorithm to solve any problem using Greedy techniques. Write any two characteristics of Greedy Algorithm.
5. Define minimum spanning tree. Apply Prim's Algorithm to find minimum cost spanning tree for the following graph :

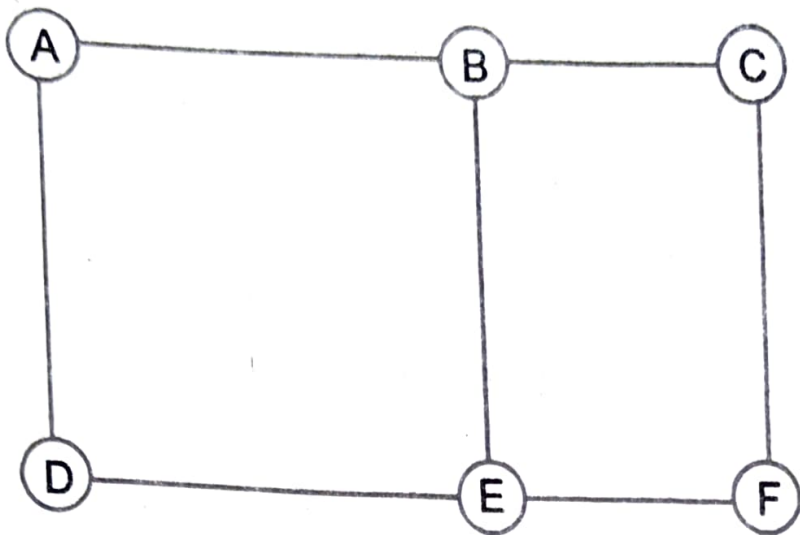


✓ 6. Explain Different Hashing Techniques with the help of an example.

7. Write recursive pre-order traversal algorithm and traverse the following tree in pre-order.



✓ 8. Differentiate between Depth-First-Search (DFS) Vs. Breath First Search (BFS) with respect to method of solving a problem and time complexity. For the following graph write DFS (sequence of traversal) from the node A:



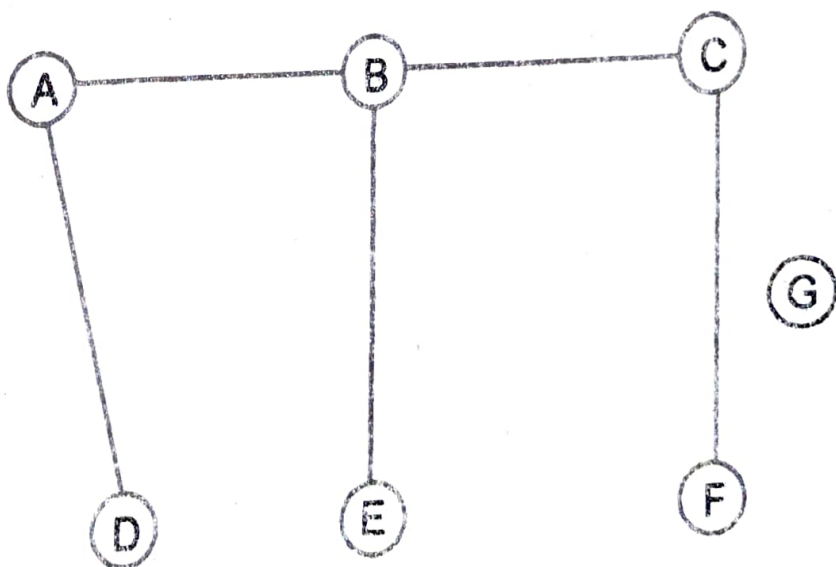
Group – B

Answer all the questions :

3×10 = 30

9. Explain Euclid's Algorithm to find the GCD of two integers with an example.
10. Solve the following Recurrence using Recursion tree method : $T(n) = 2T(n-1) + 1$.
11. Write time complexities of (i) Binary search and (ii) Merge sort.
12. Mention Best Case, Average Case and Worst Case complexities of linear search. Arrange following complexities in ascending order : $O(n \log n)$, $O(n^3)$, $O(n)$, $O(n^2)$, $O(1)$.

13. List any four examples of problems using Greedy Technique.
14. Differentiate between Prim's and Kruskal's Algorithm.
15. Define the Term :
- (i) Binary Tree
 - (ii) Complete Binary Tree
 - (iii) BST
16. Explain any three operations on set.
17. Find adjacency matrix of following graph :



18. Apply master method to solve the following recurrence relation :

(i) $T(n) = 2T(n/2) + n$

(ii) $T(n) = 3T(n/4) + n \log n$



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Group – A

Answer any **four** questions of the following :

$10 \times 4 = 40$

- ✓ 1. Write and explain the file system of Linux.
- ✓ 2. Write Ten(10) Commands of Linux with example.
- ✓ 3. What is Shell ? Write all types of shell.
4. Write a shell program to find the factorial of a number.

5. Write any shell program related to array.

✓ 6. Write and explain the role of System Administrator.

7. Write about the function of System Manager.

8. Discuss the following :

(a) AWK

(b) File Handling

Group – B

Answer all questions of the following :

3×10 = 30

✓ 9. What is Inode numbers ?

✓ 10. What is files and directories ?

✓ 11. What is Vi-editor ?

✓ 12. Write the syntax of case statements.

✓ 13. What is Strings ?

✓ 14. Write about Linux Utilities.

- ✓ 15. What is Grep and Sed ?
- ✓ 16. What is command line ? Explain.
- ✓ 17. How to create a file in Linux ?
- ✓ 18. What is Kernel ?



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Group – A

Answer any four questions : $10 \times 4 = 40$

- ✓ 1. What is error ? Write and explain all types of error.
- ✓ 2. Find the value of $f(8)$ by Newton's divided difference formula

x	$f(x)$
4	48
5	100
7	294
10	900
11	1210
13	2028

2. Find $f(46)$ of :

Age x	Premium $f(x)$
45	2.871
50	2.404
55	2.083
60	1.862
65	1.712

4. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by

Simpson's 1/3rd rule and Trapezoidal rule.

5. Solve : $y' = xy$ for $x = 1.4$

Initially $x = 1, y = 2$ ($h = .2$) using R. K. 4th order.

✓ 6. Solve by Gauss elimination method :

$$2x - 6y + 8z = 24$$

$$5x + 4y - 3z = 2$$

$$3x + y + 2z = 16.$$

✓ 7. Find the real root of $f(x) = x^3 - x - 1 = 0$.

8. Solve by Jacobi Iteration method :

$$83x + 11y + 4z = 95$$

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71.$$

Group – B

Answer all questions :

$3 \times 10 = 30$

✓ 9. What is general error formula ?

10. What is iteration method ?

11. Write about finite difference.

✓ 12. What is Lagrange's interpolation method ?

13. What is Taylor's series ?
14. Write about Matrix Inversion Method ?
15. What is interpolation ?
16. What is method of false position ?
17. Write the properties of Divided differences ?
18. Write about Numerical analysis ?



(16) Put $a, 1, 2$ — and check its nature.

$$x = \frac{aF(b) - b(F(a))}{F(b) - F(a)}$$

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Group – A

Answer any **four** questions : $10 \times 4 = 40$

1. What are the objectives of Testing ? What are the criterion for stopping the testing ?
- ✓ 2. Explain different stages of a bug life cycle with a neat diagram.
3. Explain Equivalence class Partitioning and

Boundary Value Analysis used in Black Box Testing.

4. Explain Control Structure Testing method with suitable examples.
- ✓ 5. What is SDLC ? Explain different phases in SDLC ?
6. Explain the different testing levels used in software development.
- ✓ 7. Why do we need test cases ? What are the various fields of a Test Case.
8. Explain Defect Reporting Format with a neat and clean diagram.

Group – B

Answer all questions :

3×10 = 30

9. Differentiate between Retesting and Regression Testing.
- ✓ 10. Is the user involved in Testing ? If so, in which type of Testing ?

11. What is Cyclomatic Complexity ? How we find the complexity of any program ?
- ✓ 12. Differentiate between QA and QC.
- ✓ 13. What is the difference between Test Planning and Test Strategy ?
- ✓ 14. What are the advantages of Agile Methodology over other traditional development methods ?
- ✓ 15. Explain the difference between Black Box Testing and White Box Testing.
- ✓ 16. What is Performance Testing ? Why is it useful ?
17. What is Ad-hoc testing ?
- ✓ 18. What's the most important characteristics of a Tester ?

